Software Requirements Specification (SRS)

For DormFlow Meals – Dormitory & Meal Management System

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Document Control

Version	Date	Author	Description of Changes
1.0	2026-03-15	[Your Name]	Initial draft
2.0	2026-03-17	[Your Name]	Added visual design, database schema, API specs, dashboard details, feature matrix, and future roadmap.

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1. Introduction

1.1 Purpose

This document specifies the requirements for DormFlow Meals, a web-based management system for dormitories that integrates traditional dorm operations (room allocation, rent collection, maintenance) with comprehensive meal management features. The system serves three primary user groups: Landlords/Administrators who manage one or multiple dormitories, Tenants/Students who reside in them, and Chefs who oversee meal preparation.

1.2 Scope

The product will provide:

- Landlord Dashboard: Manage multiple dorms, rooms, tenants, payments, maintenance tickets, and meal operations (meal plans, automated counting, cost calculation).
- Tenant Dashboard: View personal information, room details, rent and meal bills, submit maintenance requests, and control meal preferences (ON/OFF toggle).
- Chef Dashboard: View real-time meal counts, access weekly meal plans, and receive notifications.
- Meal Management: Weekly meal planning, real-time meal counting, time-controlled meal toggles, auto-assigning meals, and meal cost distribution.

- Subscription Plans: Tiered access (Starter, Growth, Pro) with varying limits and features.
- Data Analytics: Charts and reports for financial trends, occupancy, meal consumption, and cost breakdowns.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Admin	Landlord or dormitory manager with full control.
Tenant	Student or resident living in the dorm.
Chef	Staff member responsible for meal preparation; can view meal counts and plans.
Meal Toggle	Feature allowing tenants to turn meals ON/OFF for specific dates.
Meal Counter	Automatic calculation of total meals per meal type per day.
SRS	Software Requirements Specification.
SaaS	Software as a Service.
RBAC	Role-Based Access Control.
RLS	Row Level Security.

1.4 References

- Original DormFlow project proposal (PDF)
- Dormefy.com (inspiration for meal management features)
- Material Design 3 guidelines
- Tailwind CSS documentation

1.5 Overview

The remainder of this document describes the overall product, its features, user interfaces, data analysis, subscription model, and technical requirements.

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2. Overall Description

2.1 Product Perspective

DormFlow Meals is a new, standalone SaaS product. It replaces manual processes (spreadsheets, paper logs, messaging) with an integrated platform. It will be built as a responsive web application using modern technologies (e.g., Next.js, Supabase).

2.2 User Characteristics

User Role	Description	Key Needs
Landlord (Admin)	Owns/manages one or more dorms.	Oversee operations, track payments, manage meals, generate reports.
Tenant	Student living in the dorm.	View rent/meal bills, control meals, request maintenance.

Chef	Prepares meals; may be a tenant or staff.	See daily meal counts, view meal plans, adjust cooking quantities.

2.3 Assumptions and Dependencies

- Users have internet access and a modern web browser.
- The system will integrate with a payment gateway (e.g., ABA PayWay) for online payments (optional initially).
- Email/SMS notifications will be used for alerts.
- Data will be stored securely in a cloud database.

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3. System Architecture

3.1 High-Level Architecture

```text
┌──────────────────┐      ┌──────────────────┐      ┌──────────────────┐
│   Next.js        │──────►│   Supabase       │──────►│   PostgreSQL     │
│   Frontend       │      │   Backend        │      │   Database       │
└──────────────────┘      └──────────────────┘      └──────────────────┘
         │                         │                         │
         │                         │                         │
         ▼                         ▼                         ▼
┌──────────────────┐      ┌──────────────────┐      ┌──────────────────┐
│   Vercel         │      │   Supabase       │      │   Row Level      │
│   Hosting        │      │   Storage        │      │   Security       │
└──────────────────┘      └──────────────────┘      └──────────────────┘
```

3.2 Technology Stack

Layer	Technology	Justification
Frontend	Next.js 14 (App Router)	Server components, API routes, optimal performance
Styling	Tailwind CSS + DaisyUI	Rapid development, consistent design
State Management	Zustand / React Context	Lightweight, scalable
Charts	Recharts	Beautiful, responsive data visualization
Backend	Next.js API Routes	Integrated with frontend, serverless
Database	Supabase (PostgreSQL)	Real-time capabilities, RLS, authentication
Authentication	Supabase Auth	Email/password, Google OAuth, magic links
File Storage	Supabase Storage	Secure file uploads (receipts, images)
Real-time	Supabase Realtime	Live meal count updates
Payments	ABA PayWay / Wing API	Local payment integration

Hosting	Vercel	Optimized for Next.js, automatic deployments

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4. User Roles and Permissions

Permission	Admin	Tenant	Chef
Manage multiple dorms	✅ (Pro/Premium)	❌	❌
Create/edit rooms	✅	❌	❌
Assign tenants	✅	❌	❌
Create meal plans	✅	❌	❌
View meal counts	✅	Limited (own)	✅ (all)
Toggle meals	❌	✅	❌
View own invoices	❌	✅	❌
View all invoices	✅	❌	❌
Submit maintenance	❌	✅	❌
View all maintenance	✅	Limited (own)	❌

Access financial reports	✅ (Pro/Premium)	❌	❌
Manage chefs	✅ (Premium)	❌	❌
System notifications	✅	✅	✅

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5. Functional Requirements

5.1 User Authentication & Roles

- FR1.1 The system shall allow users to register and log in using email/password or Google OAuth.
- FR1.2 Users shall have one of three roles: `admin`, `tenant`, or `chef`. A user can be a tenant in one dorm and a chef in another.
- FR1.3 Admins can create/manage multiple dormitories from a single account.
- FR1.4 Admins can invite tenants and chefs via email; invited users receive a link to set their password.

5.2 Dormitory Management

- FR2.1 Admins can create dormitory profiles with name, address, contact info, and settings (meal cost calculation method, billing cycle).
- FR2.2 Admins can view a list of all dorms with key stats: total rooms, occupancy rate, pending payments, today's meal count.
- FR2.3 Admins can assign managers (other admin accounts) to specific dorms (for multi-admin scenarios).

5.3 Room/Seat Allocation & Rent

- FR3.1 Admins can define rooms within a dorm, each with: room number, capacity, monthly rent, availability.

- FR3.2 Tenants can be assigned to specific seats/rooms with start and end dates.
- FR3.3 The system shall track rent due dates and generate monthly invoices automatically.
- FR3.4 Tenants can view their rent history and make payments (cash or online). Admins can mark payments as received.
- FR3.5 A public directory of available seats can be enabled by the admin, allowing prospective tenants to submit booking requests.

5.4 Meal Management

- FR4.1 Weekly Meal Planner: Admins/chefs can create meal plans for each week (breakfast, lunch, dinner menu). Tenants can view the plan.
- FR4.2 Meal Toggle (ON/OFF):
    - Tenants can turn meals ON/OFF for future dates up to a configurable cutoff time.
    - Changes are recorded and instantly reflected in meal counts.
- FR4.3 Auto Meal Assigning: Admins can set a default meal preference for all tenants (e.g., all meals ON for the next quarter) that applies unless toggled off.
- FR4.4 Automated Meal Counter:
    - The system calculates daily meal counts in real-time based on tenant toggles.
    - Counts are displayed to the admin/chef for each meal type.
- FR4.5 Time-Controlled Counter: Admins can set cut-off times for each meal. After cut-off, toggles are locked for that meal.
- FR4.6 Meal Cost Calculation:
    - At the end of each billing period, the system calculates each tenant's meal cost based on:
        - Number of meals taken × per-meal rate, OR
        - Shared total food cost divided by total meals taken (proportional).
    - Admins configure the calculation method in dorm settings.
    - Meal costs are added to the tenant's monthly invoice.
- FR4.7 Meal Reports: Tenants can view their personal meal history and projected cost. Admins can download summaries for the dorm.

5.5 Payment Tracking

- FR5.1 Rent and meal costs are combined into a single monthly invoice per tenant.
- FR5.2 Tenants can view invoices, see breakdown (rent + meals + adjustments), and make payments.
- FR5.3 Admins can record cash payments and mark invoices as paid.
- FR5.4 The system shall send reminders for overdue payments.

5.6 Maintenance Requests

- FR6.1 Tenants can submit maintenance tickets with description, category (plumbing, electrical, etc.), and optional photo.
- FR6.2 Admins receive notifications and can update ticket status (open, in progress, resolved).
- FR6.3 Tenants can track the status of their requests.

5.7 Reports & Analytics

- FR7.1 Admin dashboard includes charts:
    - Financial: income vs. expenses (maintenance, food) over time.
    - Occupancy: pie chart of occupied vs. available seats.
    - Meal consumption: line chart of meals taken per day/week.
    - Meal cost breakdown: pie chart of food cost vs. other expenses.
- FR7.2 Tenants see personal progress: payment history, meal consumption chart.

5.8 Subscription & Billing (SaaS)

- FR8.1 The platform uses a subscription model for admins. Tenants and chefs use the system for free.
- FR8.2 Admins can subscribe to a plan via the application; payment processed through integrated gateway.
- FR8.3 The system enforces limits based on the active plan (e.g., max rooms, meal features enabled).

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6. Non-Functional Requirements

6.1 Performance

- Page load time < 2 seconds for typical operations.
- Real-time meal counts update within 2 seconds of a tenant toggle.
- Support up to 1,000 concurrent users initially.

6.2 Security

- All data transmitted over HTTPS.
- Passwords hashed using bcrypt.
- Role-based access control (RBAC) enforced at API level.
- Row-level security (RLS) in database.

6.3 Usability

- Mobile-responsive design (mobile-first).
- Intuitive navigation with clear labels.
- Support for Khmer language (phase 2).

6.4 Reliability

- 99.5% uptime during peak hours (evenings).
- Automated daily backups.
- Graceful degradation when services unavailable.

6.5 Scalability

- Architecture supports horizontal scaling.
- Database indexing for fast queries on meal counts and invoices.
- Caching strategy for frequently accessed data.

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7. UI/UX Design Specifications

7.1 Design Philosophy

"Clarity meets elegance" – The interface recedes, letting data and actions take center stage. Every screen is designed to be immediately understandable while feeling premium and delightful.

7.2 Color Palette

Color	Hex Code	Usage
Primary Blue	`#2563EB`	Primary actions, links, active states
Blue Light	`#3B82F6`	Hover states, secondary accents
Blue Dark	`#1D4ED8`	Pressed states, important highlights
Pure White	`#FFFFFF`	Dashboard backgrounds, cards, modals
Off-White	`#F8FAFC`	Page backgrounds, subtle contrast
Black	`#0F172A`	Primary text, headings
Gray 700	`#334155`	Secondary text, labels
Gray 400	`#94A3B8`	Disabled states, placeholder
Gray 200	`#E2E8F0`	Borders, dividers
Success Green	`#10B981`	ON toggles, paid status

Warning Orange	`#F59E0B`	OFF toggles, pending
Error Red	`#EF4444`	Overdue, critical alerts

7.3 Typography

- Primary Font: `Inter` – Clean, modern, highly readable.
- Khmer Font: `Kantumruy Pro` – For Khmer language support (phase 2).
- Headings: Inter Bold, 24/20/18/16px.
- Body: Inter Regular, 16/14px.
- Labels: Inter Medium, 14px.
- Numbers: Inter Mono (for financial data).

7.4 Iconography

- Icon Library: Heroicons (outline style) – Consistent, modern.
- Size: 20px default, 24px for navigation.
- Color: Inherits text color or primary blue for actions.

7.5 Component Library

- Buttons: Primary (blue fill), secondary (white, blue border), ghost, danger.
- Cards: White background, subtle shadow (`0 1px 3px rgba(0,0,0,0.05)`), rounded-xl.
- Inputs: White background, gray border, rounded-lg, focus: blue ring.
- Tables: Clean, minimal borders, hover states, sticky headers.
- Modals: Centered, white background, rounded-2xl, overlay blur.
- Navigation Sidebar: White with blue active indicators.

7.6 Responsive Behavior

- Mobile (<640px): Single column, bottom navigation for dashboards.
- Tablet (640px–1024px): Two-column layouts where appropriate.
- Desktop (≥1024px): Full multi-column layouts with sidebar navigation.

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8. Dashboard Specifications

8.1 Admin Dashboard

Overview Tab (All Tiers)

Section	Components	Data Source
Stats Cards	Total rooms, occupied, vacant, pending payments, today's meal counts	Real-time queries
Quick Actions	Add Tenant, Record Payment, Create Meal Plan	Modal forms
Recent Activity	List of recent tenant toggles, payments, maintenance	`notifications` table

Pro & Premium Only

- Financial Chart: Bar chart – income vs. expenses (last 6 months)
- Occupancy Chart: Pie chart – occupied vs. vacant
- Meal Consumption Chart: Line chart – meals per day/week

Full Menu (Premium)

- Dorms (multi-dorm management)
- Rooms (unlimited)
- Tenants (advanced tools)
- Meal Management (full suite)
- Payments (full financial)

- Maintenance (priority queue)
- Chefs (manage kitchen team)
- Reports (advanced analytics)
- Settings (custom branding, API)

8.2 Tenant Dashboard

Tab	Components	Description
Home	Room info, rent due, meal summary, quick actions	Overview of personal status
My Meals	Weekly meal plan, toggle buttons for future meals, personal meal count	Control and view meals
Invoices	List of invoices with status, download link	Payment history
Maintenance	Submit request, track existing tickets	Issue reporting
Profile	Update contact info, preferences	Account management

8.3 Chef Dashboard

Section	Components	Description
Today's Counts	Large cards: breakfast/lunch/dinner numbers	Real-time kitchen view

Meal Plan	Weekly menu display	Reference for cooking
Notifications	Alerts for significant toggle changes	Stay updated
Counts History	Simple chart of past meal counts	Trend spotting

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9. Subscription Model (SaaS)

9.1 Tiered Plans

Tier	Price	Max Rooms	Meal Features	Reports	Multi-Dorm	Chef Mgmt	Best For
Starter	Free	3	Basic ON/OFF only	❌	❌	❌	Small dorms, trial
Growth	$15/mo	20	Full planner, real-time counters, cost calculation	Basic	❌	❌	Growing dorms with meals

Pro	$45/mo	Unlimited	Advanced analytics, forecasting, wastage tracking	Advanced + scheduled	✅	✅	Large dorms, multiple properties

Annual plans offer a 20% discount.

9.2 Feature Availability Matrix

Feature	Starter	Growth	Pro
Max Rooms	3	20	Unlimited
Multiple Dorms	❌	❌	✅
Room Management	✅	✅	✅
Tenant Management	Basic	Full	Advanced
Basic Meal ON/OFF	✅	✅	✅
Weekly Meal Planner	❌	✅	✅
Real-time Meal Counters	❌	✅	✅
Meal Cost Calculation	❌	✅	✅
Auto Meal Assigning	❌	✅	✅

Chef Role	❌	❌	✅
Payment Tracking	Manual	Automated	Full Suite
Maintenance	Basic	Full	Priority + Analytics
Basic Reports	❌	✅	✅
Advanced Analytics	❌	❌	✅
API Access	❌	❌	✅
Custom Branding	❌	❌	✅

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10. Data Analysis & Business Intelligence

Metric	Description	Visualization
Total Income vs Expenses	Monthly comparison of rent collected vs. maintenance/food costs	Bar chart (last 6 months)
Occupancy Rate	Percentage of occupied seats per dorm	Pie chart
Meal Consumption Trend	Daily/weekly meals taken by type	Line chart

Tenant Meal Preferences	Most common meal toggle patterns	Bar chart
Payment Overdue	List of tenants with pending payments	Table with alerts

All charts are interactive and can be exported (Pro+).

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11. API Specifications

11.1 Endpoint Overview

Endpoint	Method	Purpose	Auth
`/api/auth/login`	POST	User login	Public
`/api/auth/register`	POST	User registration	Public
`/api/auth/logout`	POST	Logout	User
`/api/dorms`	GET	List user's dorms	Admin
`/api/dorms`	POST	Create new dorm	Admin
`/api/rooms`	GET	List rooms in a dorm	Admin/Tenant
`/api/rooms`	POST	Add room	Admin

`/api/tenants`	GET	List tenants	Admin
`/api/tenants`	POST	Assign tenant	Admin
`/api/meal-plans`	GET	Get meal plans	All
`/api/meal-plans`	POST	Create/update meal plan	Admin/Chef
`/api/meal-toggles`	GET	Get user's toggles	Tenant
`/api/meal-toggles`	POST	Toggle meal	Tenant
`/api/meal-counts`	GET	Get real-time counts	Admin/Chef
`/api/invoices`	GET	List invoices	Admin/Tenant
`/api/maintenance`	GET	List tickets	Admin/Tenant
`/api/maintenance`	POST	Create ticket	Tenant
`/api/notifications`	GET	Get user notifications	All
`/api/subscription`	GET	Current plan details	Admin
`/api/subscription/upgrade`	POST	Upgrade/downgrade	Admin